

Derivative Applications and Modeling

Curve analysis with the first and second derivative, related rates from one changing quantity, and optimization with a domain check

Before you start.

- The problems are deliberately mixed. Decide first *which* question is being asked: a rate that is changing in time (related rates), a largest or smallest value (optimization), or the shape of a graph (curve analysis).
- In a related-rates problem, differentiate the relationship with respect to t *before* substituting any numbers. Substituting first freezes a variable that is not constant.
- In an optimization problem, an answer is not finished until you have checked the endpoints of the physical domain and rejected the roots that lie outside it.
- Show the justification, not just the number. On an AP free response the sign chart or the second-derivative value is where the points are.

1. Let $f(x) = x^3 - 6x^2 + 9x + 2$. Find $f'(x)$.

Answer: _____

2. Using the same $f(x) = x^3 - 6x^2 + 9x + 2$, find every critical number of f , and state on which intervals f is increasing.

Answer: _____

3. A spherical weather balloon is inflated so that its radius grows at a constant 2 centimetres per second. Its volume is $V = \frac{4}{3}\pi r^3$.

Find $\frac{dV}{dt}$ at the instant the radius is 5 centimetres. Give an exact answer in terms of π (cubic centimetres per second).

Answer: _____

4. A rectangular garden is fenced on three sides; the fourth side is an existing wall. There are 60 metres of fence available. If x is the length of each of the two sides perpendicular to the wall, the enclosed area is $A(x) = x(60 - 2x)$.

(a) Find the value of x that maximizes A .

Answer: _____

(b) Find the maximum area, in square metres.

Answer: _____

5. Let $f(x) = x^4 - 4x^3$. Find every x at which $f''(x) = 0$, then decide which of them are inflection points of f and justify your decision.

Answer: _____

6. A 13 foot ladder leans against a vertical wall. The foot of the ladder is pulled away from the wall at a constant 4 feet per second.

How fast is the top of the ladder sliding down the wall at the instant the foot is 5 feet from the wall? A negative answer is expected — say what the sign means.

Answer: _____ ft/s

7. An open-topped box is made from a square sheet of cardboard 12 inches on a side by cutting a square of side x from each corner and folding up the flaps. Its volume is $V(x) = x(12 - 2x)^2$, and physically $0 < x < 6$.

(a) Solve $V'(x) = 0$. Report both roots, then say which one the domain rejects and why.

Answer: _____

(b) Find the maximum volume, in cubic inches.

Answer: _____

8. Let $f(x) = x^3 - 3x^2 - 9x + 5$ on the closed interval $[0, 4]$.

(a) Find every critical number of f , before the interval is imposed.

Answer: _____

(b) Find the absolute maximum value and the absolute minimum value of f on $[0, 4]$, and say where each occurs.

Answer: _____

9. Water is poured into an inverted right circular cone at a constant 9 cubic feet per minute. The cone's radius is always half its water depth, so $r = h/2$ and $V = \frac{\pi h^3}{12}$.

Find $\frac{dh}{dt}$ at the instant the depth is 6 feet. Give an exact answer in terms of π .

Answer: _____ ft/min

10. Synthesis. A rectangle has its base on the x -axis and its two upper corners on the parabola $y = 9 - x^2$. If the upper right corner is $(x, 9 - x^2)$ with $0 \leq x \leq 3$, the rectangle's area is $A(x) = 2x(9 - x^2)$.

- (a) Solve $A'(x) = 0$ and report both roots.

Answer: _____

- (b) Justify that the root you kept gives the *absolute* maximum area on $[0, 3]$ — not merely a critical point — and explain why a table of a few values of A could never settle that question.